LESSON 2.6

Divide Multi-Digit Numbers Using an Algorithm

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

WHY THIS MATTERS The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation. How many bars does each section receive?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Algorithm *Spanish: Algoritmo*	A step-by-step set of rules or instructions to solve a math problem.	Long Division Algorithm — D (Divide), M (Multiply), S (Subtract), B (Bring Down)
Dividend *Spanish: Dividendo*	The total number being divided into equal groups (the number inside the division bracket).	In 1,344 ÷ 12 = 112, the dividend is 1,344 — the total amount being divided
Divisor *Spanish: Divisor*	The number of equal groups you are dividing into (the number outside the division bracket).	In 1,344 ÷ 12 = 112, the divisor is 12 — the number of equal groups
Quotient *Spanish: Cociente*	The answer when you divide.	In 156 ÷ 12 = 13, the quotient is 13 — there are 13 groups
Remainder *Spanish: Residuo*	What is left over when a number does not divide evenly.	17 ÷ 5 = 3 R 2 means 3 groups of 5 with 2 left over
Divide Multi-Digit Numbers *Spanish: Dividir números de varias cifras*	Separate a multi-digit number into equal groups to find a quotient.	1,344 ÷ 12 = 112

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want 1,344 ÷ 12. The dividend is 1,344 and the divisor is 12, so 12 goes outside the bracket and 1,344 goes inside.

1. DIVIDE: 12 does not fit into 1, so I look at 13. How many 12s fit into 13? One. I write 1 above the 3.
2. MULTIPLY: 1 × 12 = 12. I write 12 underneath the 13.
3. SUBTRACT: 13 − 12 = 1. That is what is left so far.
4. BRING DOWN: I bring the next digit, 4, down next to the 1 to make 14. Now I repeat the cycle.
5. REPEAT: 14 ÷ 12 → 1 (written above the 4), 1 × 12 = 12, 14 − 12 = 2, and I bring down the last 4 to make 24.
6. REPEAT: 24 ÷ 12 → 2 (written above the last 4), 2 × 12 = 24, 24 − 24 = 0. Nothing is left to bring down, so the remainder is 0.
7. The digits I wrote are 1, 1, 2, so 1,344 ÷ 12 = 112. I check by multiplying: 12 × 112 = 1,344.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's do 384 ÷ 8. DIVIDE: 8 does not fit into 3, so we look at 38. How many 8s fit into 38? (4 — write it above the 8.)

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- 2,184 ÷ 14 = _____ because I repeated divide, multiply, subtract, and bring down until there were no digits left to bring down, and what was left at the end was _____.

Word bank Algorithm | Dividend | Divisor | Quotient | Remainder | Divide Multi-Digit Numbers | divide | multiply | subtract | bring down | repeat | quotient

WATCH OUT!

A common mistake

A common mistake in Divide Multi-Digit Numbers is skipping a placeholder zero in the quotient when a digit doesn't divide evenly, which shifts every digit after it out of place. For example, dividing 3,216 ÷ 3: 3 ÷ 3 = 1, but the next digit 2 doesn't divide by 3, so a 0 must be written before continuing

LESSON 2.6

Divide Multi-Digit Numbers Using an Algorithm

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

WHY THIS MATTERS The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation. How many bars does each section receive?

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want 1,344 ÷ 12. The dividend is 1,344 and the divisor is 12, so 12 goes outside the bracket and 1,344 goes inside.

1. DIVIDE: 12 does not fit into 1, so I look at 13. How many 12s fit into 13? One. I write 1 above the 3.
2. MULTIPLY: 1 × 12 = 12. I write 12 underneath the 13.
3. SUBTRACT: 13 − 12 = 1. That is what is left so far.
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6. REPEAT: 24 ÷ 12 → 2 (written above the last 4), 2 × 12 = 24, 24 − 24 = 0. Nothing is left to bring down, so the remainder is 0.
7. The digits I wrote are 1, 1, 2, so 1,344 ÷ 12 = 112. I check by multiplying: 12 × 112 = 1,344.

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Run the long-division cycle. Fill in the quotient.

Problem	Divide	Multiply & Subtract	Bring Down	Quotient
384 ÷ 8				
672 ÷ 6				

STEPS TO TRY

1. Read the row left to right — it is one trip around the cycle: divide, multiply, subtract, bring down.
2. The Quotient column wants the WHOLE answer, not just the first digit. Keep repeating the cycle until no digits are left to bring down.
3. Row 1: after you bring down the 4 you are looking at 64. Divide, multiply and subtract once more, then read the digits you wrote across the top.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 936 ÷ 12?

(A) 68

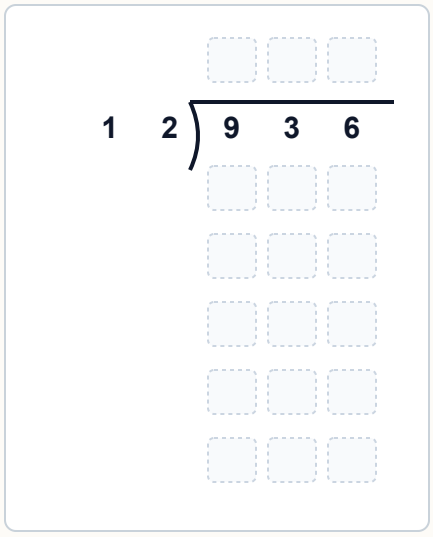
(B) 76

(C) 78

(D) 87

STEPS TO TRY

1. Estimate first: 12 × 70 = 840 and 12 × 80 = 960, so the quotient must land between 70 and 80.
2. Set it up as long division and DIVIDE first — how many 12s fit into 93?
3. DIVIDE: 93 ÷ 12 → 7. MULTIPLY: 7 × 12 = 84. SUBTRACT: 93 − 84 = 9. BRING DOWN the 6 to make 96, then repeat the cycle.



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

3 **Circle the letter of the best answer. Show how you know.**

What is 2,485 ÷ 5?

(A) 487

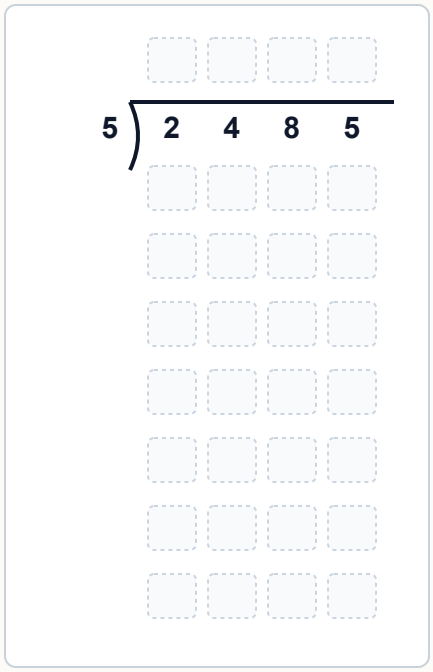
(B) 496

(C) 497

(D) 507

STEPS TO TRY

1. Size it up: 5 × 500 = 2,500, just a bit more than 2,485 — so the quotient is a little under 500.
2. Set it up as long division and DIVIDE first — how many 5s fit into 24?
3. DIVIDE: 24 ÷ 5 → 4. MULTIPLY: 4 × 5 = 20. SUBTRACT: 24 − 20 = 4. BRING DOWN the 8 to make 48, then repeat until the last digit is used.



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

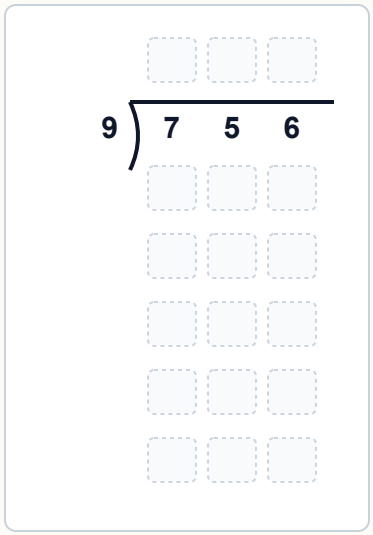
4 **Circle the letter of the best answer. Show how you know.**

What is 756 ÷ 9?

(A) 83
(B) 84
(C) 86
(D) 94

STEPS TO TRY

1. Anchor on a fact you know: 9 × 80 = 720, which is close to 756 — so the quotient is just above 80.
2. Set it up as long division and DIVIDE first — how many 9s fit into 75?
3. DIVIDE: 75 ÷ 9 → 8. MULTIPLY: 8 × 9 = 72. SUBTRACT: 75 − 72 = 3. BRING DOWN the 6 to make 36, then repeat the cycle.



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

EXPLAIN YOUR THINKING

You ran the cycle three times. What is 2,184 ÷ 14, and how did you know when to stop repeating divide, multiply, subtract, bring down?

SENTENCE STARTER 2,184 ÷ 14 = ___ because I repeated divide, multiply, subtract, and bring down until there were no digits left to bring down, and what was left at the end was ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help